

Photon-based High-dose Single-fraction Radiosurgery, an Effective Treatment Modality for Large and Posterior Uveal Melanoma

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Abstract. *Background/Aim:* Uveal melanoma (UM) is the most common primary intraocular malignant tumor. This malignancy is frequently treated using brachytherapy, stereotactic radiotherapy, or proton therapy. The objective of this study was to assess the role of stereotactic radiosurgery in the treatment of large and posterior UM. *Patients and Methods:* From January 2014 to July 2021, we treated 65 patients (median age=71 years; range=31-89 years) affected by UM. Inclusion criteria were Eastern Cooperative Oncology Group (ECOG) performance status (PS) ≤ 2 , life expectancy > 6 months, tumor thickness > 10 mm, diameter > 16 mm or posterior UM. The treatment was delivered with a True BeamTM LINAC with arc modulation technique. All patients received 27 Gy in one fraction (biological effective dose ≈ 100 Gy, assuming an α/β of 10). *Results:* The median follow-up was 36 (range=3-90) months. Acute toxicities were reported in 14 patients, whereas late toxicity occurred in 45 (69.2%). Fifteen patients (23.0%) underwent enucleation: eight (12.3%) for failure of local control and seven (10.7%) for late treatment co-morbidities. The 5-year local control, and progression-free, metastasis-free, enucleation-free, and overall survival rates were 80%, 43%, 62%, 65% and 56%, respectively. In multivariate

analysis, tumor dimensions significantly influenced survival [larger basal diameter: progression-free [hazard ratio (HR)=2.42] and overall (HR=2.61) survival; greater thickness: overall survival (HR=2.36)]. In multivariate analysis, patients without local control had a higher risk of distant metastasis (HR=3.25). *Conclusion:* Stereotactic radiosurgery offers an effective and safe approach for selected cases of UM due to the satisfactory results in terms of local control, eye conservation and survival.

Uveal melanoma (UM), a rare malignancy arising from melanocytes, is the most common primary intraocular cancer in adults, representing 3-5% of all melanomas (1-3). Between 1973 and 2008, the Surveillance, Epidemiology, and End Results database of the United States National Cancer Institute included 4,070 patients with primary UM and reported approximately five cases per million per year (3). Moreover, UM is the most aggressive intraocular tumor in adults: up to 50% of patients die within 15 years after enucleation or other therapeutic methods (4).

UM occurs predominantly in elderly men between 60 and 80 years, with a peak of incidence at 70 years and plateau after 75 years (5, 6).

Main risk factors associated with this tumor are genetic factors, race, color of the eyes, fair coloring of the skin and the ability to tan. The role of sunlight and other environmental exposures have been discussed in the literature but their influence on the pathogenesis is currently unproven. UM mainly metastasizes in other regions of the body such as liver, breast, lung, and kidney (2-4).

In the past, enucleation represented the standard therapy (7) but over the past four decades, eye-sparing treatments with radiotherapy (RT) have assumed a predominant role in

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the treatment of UM, especially in the form of brachytherapy (BT) or proton-beam RT, showing promising results and allowing globe preservation and partial vision sparing. Moreover, the Collaborative Ocular Melanoma Study, a multicenter randomized trial, found no differences in the 5-year overall survival rates after enucleation or BT in the treatment of small- or medium-size choroidal melanoma and established the role of BT in the treatment of UM (8, 9).

Currently, BT with 125-iodine and 106-ruthenium represents the first-line treatment for UM, as delivery of continuous radiation at a low dose rate minimizes subsequent radiation damage to healthy ocular tissues and therefore reduces the incidence of complications.

However, BT has limitations due to tumor size (>10 mm thickness, >16 mm diameter, *i.e.*, very large UM) or tumor location (close to the optic nerve or lesions in the posterior lobe of the eye). In these cases, other conservative treatments such as proton-beam or stereotactic photon-beam RT have been suggested (10, 11). Stereotactic body radiotherapy (SBRT) is administered using a few large fractions or a single large fraction (SRS). The hypo-fractionation approach has been widely adopted as studies *in vitro* showed that UM appears to be relatively radioresistant to low fractional RT (12, 13). The development of SBRT and SRS occurred due to technological advances in image guidance and treatment delivery techniques that allow large doses to be delivered to target tumors while minimizing the dose to surrounding normal tissues.

The aim of this retrospective study was to analyze the efficacy and safety of SRS as eye-preserving treatment in large and posterior UM unfit for BT and to determine prognostic factors for local control (LC) as well as enucleation-free survival, progression-free (PFS), metastasis-free (MTFS) and overall (OS) survival.

Patients and Methods

The present study retrospectively analyzed 65 patients with primary large and posterior UM unfit for BT and treated with SRS, between January 2014 and July 2021, after assessment at the Multidisciplinary Team of Ocular Melanoma of the Azienda Ospedaliera Universitaria Pisana.

Inclusion criteria were as follows: Eastern Cooperative Oncology Group performance status ≤ 2 , normal bone marrow function, life expectancy >6 months and target lesions (thickness >10 mm, diameter >16 mm) and/or posterior lobe tumors.

In contrast to the principles of clinical oncology practice, histological evaluation is not routinely performed in the diagnosis of intraocular neoplastic lesions. So, the diagnosis of UM was based primarily on clinical examination by the Ophthalmologists of the Ophthalmic Surgery of the Azienda Ospedaliera Universitaria Pisana, with biomicroscopy, ophthalmoscopy, and standardized (A-scan B-scan) ultrasonography. Other examinations such as color fundus photography, fundus fluorescein angiography, indocyanine green angiography and optical coherence tomography were used to

confirm diagnosis. Tumor size was assessed by standardized echography. Magnetic resonance imaging and computed tomography (CT) were performed to determine tumor stage, based on the eighth edition of American Joint Committee on Cancer-Union International for Cancer Control classification for UM (14) and to exclude distant metastases at the time of diagnosis.

Local peribulbar anesthesia was performed in patients with poor compliance, whereas in the remaining cases, a fixation system was used in order to ensure a satisfactory reproducibility of the eye position compared to the simulation CT images. Our device consisted of a blinking light diode on which the patient focused their gaze. A camera system was used to monitor patient movement.

A CT simulation with a LightSpeed RT 16 CT Simulator (General Electric Co. Boston, MA, USA) was performed for all patients before treatment using a customized immobilization thermoplastic mask. CT images were acquired with 1.25 mm slice thickness in supine position for a better delineation of the clinical target volume and of the surrounding normal tissues and then transmitted to the planning system. The treatment program was to deliver $\geq 95\%$ of the prescribed dose to the planning target volume. The target delineation was contoured by a radiation oncologist according to International Commission on Radiation Units and Measurements Report 91 (15). The planning target volume was generated by adding a margin in all directions to the respective clinical target volume (median=3 mm, range=2-5 mm). Normal structures were contoured to calculate a dose-volume histogram. The treatment was delivered with a True Beam™ LINAC (Varian Medical Systems, Palo Alto, CA, USA), and volumetric-modulated arc therapy technique, using 6-MV flattener free field beams.

All the lesions were treated with one single fraction of 27 Gy (biological effective dose (BED) ≈ 100 Gy), assuming an α/β of 10. The BED was calculated using the following equation, considering 10 Gy as the α/β ratio (dose at which the linear and quadratic components of cell kill were equal) (16): $BED = n \times d (1 + d/\alpha/\beta)$, in which n =number of treatment fractions, d =dose per fraction (Gy).

Personalized immobilization, Real-time Position Management™ techniques (Varian Medical Systems) to reduce organ motion effects, and image-guided cone beam CT were used to improve treatment accuracy.

Clinical evaluation was performed by Radiation Oncologists and Ophthalmologists one week after radiotherapy and within 3 months from the end of SRS in the first year, every six months for 5 years, and every 12 months subsequently.

Radiation induced toxicities were scored according to Common Terminology Criteria for Adverse Events (v 5.0) (17). Acute toxicities were defined as symptoms occurring within 3 months after treatment completion, whereas late toxicities were defined as symptoms developing after 3 months.

All the patients were observed until they died or until July 2021. The median follow-up of survivors was 36 months (range=3-90 months).

Statistical analysis. LC, PFS, OS, enucleation-free survival and MTFS were assessed as end-points using the following covariates: Patient age, sex, eye laterality, largest tumor diameter, height of the tumor, site, integrity of the Bruch membrane and tumor stage. Survival curves were calculated with the Kaplan-Meier method and log-rank test was applied to evaluate differences between curves. Covariates resulted influencing survival ($p < 0.1$) after univariate analysis were included in a Cox regression model as multivariate

Table I. Patient and tumor characteristics.

Characteristic	Value
Age, years	
Median (range)	71 (31-89)
<65 Years	23 (35.4%)
≥65 Years	42 (64.6%)
Sex, n (%)	
Male	36 (55.4%)
Female	29 (44.6%)
Eye laterality, n (%)	
Right	33 (50.8%)
Left	32 (49.2%)
Median tumor size (range), mm	
Largest basal diameter	13.2 (4.8-22.0)
Thickness	8.7 (1.8-18.5)
Tumor site, n (%)	
Posterior segment	31 (47.7%)
Other	34 (52.3%)
Integrity of the Bruch membrane, n (%)	
Yes	52 (80.0%)
No	13 (20.0%)
AJCC-UICC stage, n (%)	
1	6 (9.2%)
2	17 (26.2%)
3	25 (38.4%)
4	17 (26.2%)

AJCC-UICC: American Joint Committee on Cancer-Union for International Cancer Control.

analysis. The results of survival analysis were expressed by hazard ratio (HR) with 95% confidence interval (95% CI) and by regression coefficient. Significance was fixed at 0.05 and all analyzes were carried out with SPSS v.27 (IBM, Armonk, NY, USA).

Results

Patient characteristics are shown in Table I. All the patients were treated with SRS (one single fraction of 27 Gy).

At 2 and 5 years, LC was 88% and 80%, PFS was 67% and 43%, MTFS was 78% and 62%, enucleation-free survival was 80% and 65%, and OS was 90% and 56%, respectively.

LC. LC was assessed with a regression of tumor dimensions or a lack of progression during follow-up. An accurate ophthalmic examination at 3 months after SRS showed LC of disease in 57 (87.7%) patients. Patients with lesions with a thickness of ≤8 mm had a significantly higher probability of LC when compared to those with larger lesions (>8 mm vs. ≤8 mm: HR=5.33, 95% CI=1.04-27; $p=0.044$) (Table II).

PFS. Patients with lesions with a basal diameter of ≤13 mm had higher probability of PFS when compared with those with larger basal diameter (>13 vs. ≤13 mm: HR=2.95, 95% CI=1.33-6.53; $p=0.008$); data confirmed at multivariate

analysis (HR=2.42, 95% CI=1.03-5.65; $p=0.041$). Moreover, patients with lesions ≤8 mm had higher probability of LC when compared to those with greater thickness, with statistical significance (HR=2.59, 95% CI=1.20-5.58; $p=0.014$). Patients with stage 3-4 UM tended to have a higher risk of progression when compared with those with other stages ($p=0.076$) (Table II).

MTFS. Distant metastases were recorded in 14 patients (21.5%): four of them (6.1%) already had metastatic disease at the time of the diagnosis. There were 12 patients (18.5%) with liver metastases, one (1.5%) with liver and brain metastases, and one (1.5%) with lung metastasis. All the patients with distant disease underwent immunotherapy. Patients with lesions with basal diameter >13 mm had a trend for lower MTFS when compared with those with larger diameter (>13 vs. ≤13 mm: HR=2.50, 95% CI=1.92-6.77; $p=0.071$) and patients with lesions with thickness >8 mm had lower probability of MTFS when compared with those with higher thickness (HR=2.87, 95% CI=1.05-7.82; $p=0.039$). Patients in whom LC was not achieved had a higher risk of distant metastasis compared with patients who had LC (univariate: $p=0.030$; multivariate: $p=0.030$) (Table III).

Enucleation-free survival. Enucleation was required in 15 patients (23.0%): eight (12.3%) due to failure of LC, seven (10.7%) due to late toxicities (three eyes developed a severe unresponsive refractory neovascular glaucoma, two had severe large scleral necrosis with risk of globe perforation and two eyes developed corneal ulcers). Patients with lesions with a tumor thickness of more than 8 mm had higher probability of enucleation when compared with those with lower thickness, with a statistical significance (HR=4.21, 95% CI=1.32-13; $p=0.015$) (Table III).

OS. A total of 26 patients (40.0%) died: Seven (10.7%) due to distant progression, eight (12.3%) due to local and distant progression, and the other 11 (17.0%) due to age/comorbidities.

Patients with lesions with basal diameter ≤13 mm had higher probability of OS when compared with those with larger diameter (>13 vs. ≤13 mm: HR=3.24, 95% CI=1.39-7.52; $p=0.006$), data confirmed at multivariate analysis (HR=2.61, 95% CI=1.09-6.21; $p=0.030$). Moreover, patients with lesions ≤8 mm thick had higher OS probability when compared with those with higher thickness, with statistical significance (>8 vs. ≤8 mm: HR=2.97, 95% CI=1.31-6.73; $p=0.009$), and confirmed at multivariate analysis (HR=2.36, 95% CI=1.01-5.50; $p=0.045$).

Patients with stage 3-4 UM had a high risk of death when compared with those with other stages ($p=0.048$), and patients with lesions of the posterior ocular segment tended to have a worse prognosis when compared with those with lesions in other sites ($p=0.081$) (Table IV).

Table II. Univariate and multivariate analysis of local control (LC) and progression-free survival (PFS).

		LC			PFS		
Factor	Comparison	HR	95% CI	<i>p</i> -Value	HR	95% CI	<i>p</i> -Value
Univariate							
Age	>70 <i>vs.</i> ≤70 Years	0.389	0.078-1.936	0.249	1.208	0.573-2.544	0.619
Sex	Female <i>vs.</i> male	1.793	0.428-7.512	0.425	1.123	0.534-2.364	0.760
Side	Right <i>vs.</i> left	0.354	0.071-1.763	0.205	0.989	0.472-2.072	0.976
Site	Posterior <i>vs.</i> other	0.743	0.123-3.018	0.678	0.560	0.267-1.176	0.126
Largest basal diameter	>13 <i>vs.</i> ≤13 mm	2.405	0.569-10	0.233	2.950	1.331-6.538	0.008
Thickness	>8 <i>vs.</i> ≤8 mm	5.336	1.049-27	0.044	2.599	1.209-5.586	0.014
Stage	3+4 <i>vs.</i> 1+2	2.167	0.433-11	0.347	2.164	0.922-5.080	0.076
Bruch’s membrane	Positive <i>vs.</i> negative	2.318	0.552-9.732	0.251	0.537	0.185-1.560	0.253
Multivariate							
Largest basal diameter	>13 <i>vs.</i> ≤13 mm	2.405	0.569-10.0	0.233	2.420	1.037-5.650	0.041
Thickness	>8 <i>vs.</i> ≤8 mm	2.167	0.433-11.0	0.347	2.069	0.740-5.783	0.165
Stage	3+4 <i>vs.</i> 1+2	2.318	0.552-9.73	0.251	0.966	0.298-3.129	0.954

CI: Confidence interval; HR: hazard ratio. Statistically significant p-values are shown in bold.

Table III. Univariate and multivariate analysis of metastasis-free survival (MTFS) enucleation-free survival (EFS).

		MTFS			EFS		
Factor	Comparison	HR	95% CI	<i>p</i> -Value	HR	95% CI	<i>p</i> -Value
Univariate							
Age	>70 vs. ≤70 Years	0.785	0.298-2.069	0.624	1.261	0.456-3.487	0.655
Sex	Female vs. male	1.144	0.429-2.896	0.824	1.175	0.425-3.247	0.756
Side	Right vs. left	1.058	0.405-2.705	0.909	0.571	0.176-1.519	0.230
Site	Posterior vs. other	0.451	0.166-1.222	0.117	0.560	0.199-1.582	0.247
Largest basal diameter	>13 vs. ≤13 mm	2.501	1.923-6.771	0.071	1.515	0.547-4.201	0.424
Thickness	>8 vs. ≤8 mm	2.872	1.054-7.827	0.039	4.215	1.329-13	0.015
Stage	3+4 vs. 1+2	2.133	0.693-6.559	0.186	2.577	0.725-9.161	0.143
Bruch's membrane	Positive vs. negative	0.426	0.097-1.880	0.260	1.265	0.401-3.992	0.688
Local control	No vs. yes	3.263	1.123-9.482	0.030	1.26	2.32-3.37	0.756
Multivariate							
Thickness	>8 vs. ≤8 mm	2.406	0.762-7.53	0.251	1.308	0.472-2.73	0.619
Local control	No vs. yes	3.252	1.119-9.451	0.030			

CI: Confidence interval; HR: hazard ratio. Statistically significant p-values are shown in bold.

SRS toxicity. Overall acute and late toxicities are reported in Table V.

Acute toxicity up to 3 months after radiotherapy was found in 14 patients (21.5%). The most common grade 1 to 2 acute adverse event was conjunctivitis (n=4, 6.0%), followed by eye pain and watering eyes (n=3, 4.5%, both). No acute grade 3-4 toxicity occurred.

Late toxicity occurred in 45 patients (69.2%). The most common grade 1 to 2 late adverse events were radiation-

induced retinopathy and cataract (n=8, 12.0% both), followed by neovascular glaucoma (n=5, 7.5%). The most common grade 3 to 4 late adverse events were exudative retinal detachment and radiation-induced retinopathy (n=6, 9.0% both), followed by neovascular glaucoma (n=5, 7.5%).

Final visual acuity was <20/200 in 37 (57.0%) patients; a progressive reduction of at least three Snellen lines was recorded in 35 (53.8%) patients.

Table IV. Univariate and multivariate analysis of overall survival.

Factor	Comparison	HR	95% CI	p-Value
Univariate				
Age	>70 vs. ≤70 Years	1.911	0.862-4.237	0.111
Sex	Female vs. male	1.177	0.541-2.559	0.681
Side	Right vs. left	1.226	0.557-2.699	0.612
Site	Posterior vs. other	0.492	0.222-1.091	0.081
Largest basal diameter	>13 vs. ≤13 mm	3.241	1.396-7.527	0.006
Thickness	>8 vs. ≤8 mm	2.979	1.318-6.735	0.009
Stage	3+4 vs. 1+2	2.681	1.009-7.152	0.048
Bruch's membrane	Positive vs. negative	0.478	0.164-1.394	0.177
Multivariate				
Largest basal diameter	>13 vs. ≤13 mm	2.611	1.097-6.215	0.030
Thickness	>8 vs. ≤8 mm	2.369	1.019-5.508	0.045
Stage	3+4 vs. 1+2	2.611	1.097-6.215	0.030

CI: Confidence interval; HR: hazard ratio. Statistically significant p-values are shown in bold.

Table V. Acute and late toxicities after radiosurgery for uveal melanoma according to Common Terminology Criteria for Adverse Events Version 5 (17).

		Grade, n (%)				
	Toxicity	1	2	3	4	Overall
Acute	Eyelid function disorder	0	1 (1.5%)	0	0	1 (1.5%)
	Uveitis	1 (1.5%)	0	0	0	1 (1.5%)
	Eye pain	5 (7.5%)	0	0	0	5 (7.5%)
	Watering eyes	3 (4.5%)	0	0	0	3 (4.5%)
	Photophobia	2 (3.0%)	0	0	0	2 (3.0%)
	Conjunctivitis	4 (6.0%)	0	0	0	4 (6.0%)
	Extraocular muscle disorder	1 (1.5%)	0	0	0	1 (1.5%)
Late	Cataract	8 (12.0%)	3 (4.5%)	3 (4.5%)	2 (3.0%)	16 (24.0%)
	Corneal ulcer	0	0	0	2 (3.0%)	2 (3.0%)
	Neovascular glaucoma	0	5 (7.5%)	5 (7.5%)	4 (6.0%)	14 (21.0%)
	Exudative retinal detachment	0	0	6 (9.0%)	3 (6.0%)	9 (15.0%)
	Radiation retinopathy	4 (6.0%)	8 (12.0%)	6 (9.0%)	0	18 (27.0%)
	Vitreous hemorrhage	3 (4.5%)	2 (3.0%)	4 (6.0%)	2 (3.0%)	11 (16.5%)
	Scleral necrosis	0	0	3 (4.5%)	2 (3.0%)	5 (7.5%)

Discussion

In recent decades, the management of patients with UM has shifted toward globe-sparing techniques, including irradiation with radioactive plaques, charged particles, and photons. Debate is still ongoing about the choice of the most favorable radiotherapy treatment schedule for large UM unfit for BT, in terms of better efficacy and minimization of adverse events. Moreover, the emergence of image-guided radiotherapy and SBRT has made permanent ablation of tumors possible with high doses of customized radiotherapy at target volumes with low morbidity.

Since 1971, several studies on the response to radiation of melanoma cell lines *in vitro*, showed a radioresistance of UM and showed that larger doses of RT fraction were required for satisfying control of this malignancy (18-21). Single doses greater than 6 Gy were needed to produce a significant effect (21). Van den Aardweg *et al.* evaluated RT response of some primary and metastatic UM cell lines and concluded that single doses of 17-20 Gy or three to four fractions of 8-10 Gy were required to eradicate a 1 cm³ tumor (22).

Zelevsky *et al.* showed that single-fraction high-dose SRS resulted in high local control for usually considered radio-resistant tumors according to classical radiobiological

ranking, such as metastatic renal cell cancer (23), and SRS was recently proposed as an alternative treatment for very large and posterior UM (20). Indeed, in their study on single-dose and fractionated stereotactic radiotherapy in the treatment of UM, Zehetmayer and coworkers did not clearly support one treatment modality over another; a single treatment is more convenient for the patient, as well as in term of reducing set up errors and interfraction variations, but increased fractionation may reduce the risk of late side-effects (24). Several reviews on SRS (mostly gamma knife-based) showed a higher rate of neovascular glaucoma (24-36%) but lower rate of optic neuropathy than with SBRT; rates of eye preservation and local control also appeared equivalent (25). A recent meta-analysis showed no difference in tumor control, survival, and toxicities, as defined in this article, between SRS and SBRT for UM (26).

Several clinical and pathological prognostic factors have been associated with poor prognosis and higher risk of metastasis for UM. These included large tumor size, posterior site, presence of orange pigment overlying the tumor, and older patient age (27, 28). Research by Kaliki *et al.* showed that the risk for metastasis and death increased three-fold with each increase in melanoma American Joint Committee on Cancer stage: 10-Year metastasis rate in posterior UM was 12% for stage I, 29% for stage II, and 61% for stage III tumors (29). In our cohort, patients with early stages of disease had a better prognosis when compared with those with higher stages (Table II and Table IV).

Tumor size (largest basal diameter and thickness) is one of the most important prognostic factors for UM (30); a meta-analysis of Diener-West *et al.* showed that the 5-year mortality rate was 16% for patients with small tumors, 32% for those with medium tumors, and 53% for those with large tumors (31). In accordance with the literature, our univariate analysis showed that large tumors conferred a poorer prognosis. Multivariate analysis confirmed the largest basal tumor diameter negatively affected PFS ($p=0.041$), and OS ($p=0.030$), whilst tumor thickness conferred poorer OS ($p=0.045$) and had a tendency for association with poorer MTFs ($p=0.084$). This is particularly important when considering that Pereira *et al.* (32) and Tarlan *et al.* (33) showed that despite modern therapies, most patients with UM develop metastases and their prognosis is generally poor.

A review by Shildkrot *et al.* showed the presence of silent liver micro-metastases even in the early stages of UM (34). In accordance with other reported data (35), our research showed that patients with local treatment failure also had a higher risk of developing metastatic disease at univariate ($p=0.030$) and multivariate ($p=0.030$) analysis. However, it is not known whether the incidence of metastasis is affected by local treatment failure or by the aggressive nature of the primary malignancy; moreover, although not definitely proven, local tumor control can reduce metastatic risk by

preventing tumor growth that might induce cytogenetic abnormalities which are more favorable for metastasis.

In accordance with literature, in our cohort, tumor location influenced UM prognosis. Melanoma of the iris has a mortality rate 5-to 10-fold lower than that of posterior UM. In a review article of >8,000 cases of UM, the 10-year metastasis rate was 33.4% for patients with ciliary body melanomas, 25% for those with choroidal melanomas and 6.9% for those with iris melanomas (36).

Regarding RT complications, Furdova *et al.* (37), as well as our study, reported grade 3-4 severe ocular comorbidities and also reported a 5-year OS and secondary enucleating rates due to radiation stereotactic comorbidities comparable to those achieved with other irradiation techniques. Furdova *et al.* in a group of 96 patients with posterior UM treated with SRS reported a LC rate of 85% at 5 years (38). Our results are slightly worse (80%), probably due to the highly unfavorable cases enrolled, or to the tumor location.

In conclusion our study suggests that SRS can be considered a non-invasive alternative treatment for selected cases of UM, unfit for conventional BT. Photon SRS is feasible, acceptably tolerated and represents an eye-preserving treatment for patients with large and unfavorably located tumors. Since melanoma is considered a radioresistant tumor, a large single dose of stereotactic RT can be considered a useful therapeutic approach for UM. Further clinical investigations are warranted to determine optimal dosing and fractionation schedules, and to properly select the subsets of patients who are likely to benefit most from this therapeutic approach.

Conflicts of Interest

The Authors did not have any conflicts of interest in regard to this study.

Authors' Contributions

Concetta Laliscia: Article drafting, text redaction and final revision; Taiusha Fuentes, Natalina Coccia and Roberto Mattioni: clinical data collection and text redaction; Federica Genovesi, Francesca Guido and Federica Cresti: text redaction and clinical data organisation; Franco Perrone: text redaction and data analysis; Riccardo Morganti: text redaction and statistical elaboration; Fabiola Paiar: text redaction and final revision.

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